

The slowest eight-letter machine

Name: _____

Date: _____

Choose a permutation of A–H: every letter appears once in each row of a substitution key. Repeat it. Maximize the first positive turn when **all eight letters** are back. Use cards and arrow loops.

1. Beat one big loop.

An eight-letter loop takes eight turns. Can two or more loops take longer? Build a key and show why its claimed return is the *first*.

My key or loops, and its return time:

2. A complete upper bound.

The return time is the least common multiple of the loop sizes. Sort by the **largest loop**. Complete the cases below; explain why no partition of eight was missed.

Largest loop	Possible remaining loops	Best return time
8	none	
7	1	
6	2 or 1+1	
5	partitions of 3	
4	partitions of 4	
At most 3	only 1, 2, 3	

3. A theorem and a next question.

Use your construction and bound to prove the optimum. Then predict the optimum for nine letters. Must adding one letter make the optimum *strictly* bigger?

My theorem, proof, and nine-letter prediction:

Try it with objects. Explain by pointing, talking, or drawing.